CLAIMS

What is claimed is:

1. A method for establishing a communication link, the method comprising the steps of:

   (a) Determining a first geographic location of a transmitting station and a second geographic location of a receiving station.

   (b) Retrieving orbital parameter data for a plurality of non-cooperative orbiting objects.

   (c) Executing instructions on a processor to perform a computational search, said search comprising:

      i. Predicting a future trajectory for each of the plurality of orbiting objects based on the retrieved orbital parameter data.

      ii. Identifying a future time and a future spatial position of a candidate orbiting object from the plurality, wherein said future spatial position satisfies a geometric condition for specular reflection of a signal from the transmitting station to the receiving station.

      iii. Generating a firing solution comprising at least the identified future time.

   (d) Transmitting a directed signal from the transmitting station at the identified future time towards the identified future spatial position, whereby the signal reflects from the candidate orbiting object and is received by the receiving station.

2. The method of claim 1, wherein the transmitting station comprises a movable antenna, and the firing solution further comprises an azimuth and an elevation for directing the movable antenna towards the future spatial position.

3. The method of claim 1, wherein the transmitting station comprises a plurality of antennas with fixed orientations, and wherein the computational search identifies a candidate

orbiting object whose future trajectory aligns with the pre-determined orientation of one of the plurality of antennas.

4. The method of claim 1, wherein the step of identifying the future spatial position further comprises calculating an angle of incidence of a vector from the transmitting station to the object and an angle of reflection of a vector from the object to the receiving station, relative to a surface normal vector at the object's position, and confirming they are equal within a specified tolerance.

5. The method of claim 1, wherein the computational search is performed by a distributed system comprising a plurality of workers, wherein each worker is assigned to evaluate a distinct future time interval.

6. The method of claim 1, further comprising a step of calculating a distance between the first and second geographic locations and selecting a subset of the plurality of orbiting objects to search based on said distance, wherein lower-altitude objects are selected for shorter distances and higher-altitude objects are selected for longer distances.

7. A system for establishing a communication link, the system comprising:

   (a) A data ingestion module configured to retrieve orbital parameter data for a plurality of non-cooperative orbiting objects from a remote data source.

   (b) A computational engine communicatively coupled to the data ingestion module, the engine comprising one or more processors configured to:

      i. Receive a first geographic location of a transmitting station and a second geographic location of a receiving station.

      ii. Predict future trajectories for the plurality of orbiting objects based on the retrieved data.

      iii. Identify a candidate orbiting object, a future time, and a future spatial position where the object will satisfy a geometric condition for specular reflection

of a signal from the transmitting station to the receiving station.

iv. Generate a firing solution comprising at least the identified future time.

(c) A transmitting station configured to receive the firing solution from the computational engine and to transmit a directed radio signal at the identified future time towards the identified future spatial position.

8. The system of claim 7, wherein the transmitting station comprises a movable antenna, and wherein the firing solution further comprises an azimuth and an elevation for aiming the antenna.

9. The system of claim 7, wherein the computational engine is further configured to filter the plurality of orbiting objects based on a kinematic property, such that only objects moving in a direction from the first geographic location towards the second geographic location are considered in the search.

10. The system of claim 7, wherein the non-cooperative orbiting objects are commercial communication satellites whose primary communication functions are not utilized by the system.